

Green Ash

Fraxinus pennsylvanica

Species background: Native to central and eastern North America, especially along rivers and floodplains, once one of the most commonly planted street trees in the US. Ash leaves and branches always grow in exact opposite pairs, not staggered -- a quick and reliable way to rule in or rule out ash on any city block.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Green Ash: Count the leaflets on a compound ash leaf and check for opposite branching.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Check my branches -- my leaves always grow in matching pairs across from each other, never staggered! Count the little leaflets that make up one big leaf.

Procedure

- 1 Engage: Ask students to find a matching pair of anything nearby.
- 2 Explore: Look at a real or pictured ash twig and check whether the leaves line up in pairs.
- 3 Explain: Explain that this matching-pairs pattern is a real clue scientists use to identify ash trees.
- 4 Art: Count the little leaflets, then draw the same number of leaf shapes in a row.
- 5 Evaluate: Student can say whether branches are 'matching pairs' or 'staggered' when shown a picture.

Discussion questions

What else grows in matching pairs in nature?

Assessment

Correctly identifies opposite (paired) branching with prompting.

Extension

Check a different tree's branches on your walk -- paired or staggered?

Green Ash: Use opposite branching and leaflet counting as identification tools.

Standards connection: NGSS 3-LS4-4; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Green ash leaflets grow in exact opposite pairs along the twig -- check if the pattern lines up. Many of the ash cousins that used to line American streets have gotten sick from a tiny invasive beetle.

Procedure

- 1 Engage: Ask what it would mean for a whole kind of tree to get sick across an entire country.
- 2 Explore: Count leaflets on a real or pictured compound ash leaf.
- 3 Explain: Introduce the emerald ash borer as an invasive threat to ash trees since 2002.
- 4 Art: Draw one compound leaf, counting and drawing each individual leaflet along the row.
- 5 Evaluate: Student counts leaflets correctly and can explain in one sentence why ash trees are threatened.

Discussion questions

Why do you think scientists are working hard to protect the ash trees that are left?

Assessment

Correct leaflet count plus a basic explanation of the threat.

Extension

Research what an emerald ash borer looks like and what its exit holes look like on bark.

Green Ash: Explain invasive pest ecology using the emerald ash borer as a case study.

Standards connection: NGSS MS-LS2-1; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, research notebook.

Teacher background

The emerald ash borer, an invasive beetle from Asia, has killed hundreds of millions of North American ash trees since 2002 by tunneling under the bark and cutting off nutrient flow.

Procedure

- 1 Engage: How might an insect from another continent end up in North America in the first place?
- 2 Explore: Research how emerald ash borer larvae damage a tree from the inside.
- 3 Explain: Discuss why native ash trees have few natural defenses against a pest they didn't evolve alongside.
- 4 Art: Illustrate the leaflets' opposite-pair pattern as a symmetrical design, then research and add small drawings of the emerald ash borer's D-shaped exit holes as a border motif.
- 5 Evaluate: Write a paragraph explaining how the borer kills a tree and why the damage is often not noticed until it's severe.

Discussion questions

What could shipping and trade regulations do to prevent the next invasive pest introduction?

Assessment

Paragraph correctly explains the borer's damage mechanism.

Extension

Research one city's response plan for managing emerald ash borer in its street trees.

Green Ash: Evaluate management strategies for an ongoing invasive pest population.

Standards connection: NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, access to research sources.

Teacher background

Emerald ash borer larvae tunnel through the phloem, cutting off nutrient transport. Research the estimated ecological and economic cost of ash mortality in North America since 2002, and evaluate one proposed management strategy -- biocontrol wasps, systemic insecticide, or preemptive removal -- for its tradeoffs.

Procedure

- 1 Engage: What tradeoffs come with each major management strategy for an invasive pest -- cost, effectiveness, and ecological side effects?
- 2 Explore: Research the estimated economic and ecological cost of ash mortality since 2002.
- 3 Explain: Evaluate one management strategy in depth, citing its documented tradeoffs.
- 4 Art: Create an infographic-style illustration mapping the emerald ash borer's spread across North America since 2002, paired with a labeled botanical sketch of the compound leaf it targets.
- 5 Evaluate: Submit the infographic and a short written evaluation of the chosen management strategy.

Discussion questions

Is it more effective to invest in protecting existing ash trees or in diversifying future street tree plantings? Why not both, and what limits that?

Assessment

Rubric covering research depth, infographic clarity, and quality of tradeoff evaluation.

Extension

Compare the ash borer crisis to Dutch elm disease -- what did (or didn't) cities learn between the two epidemics?

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.